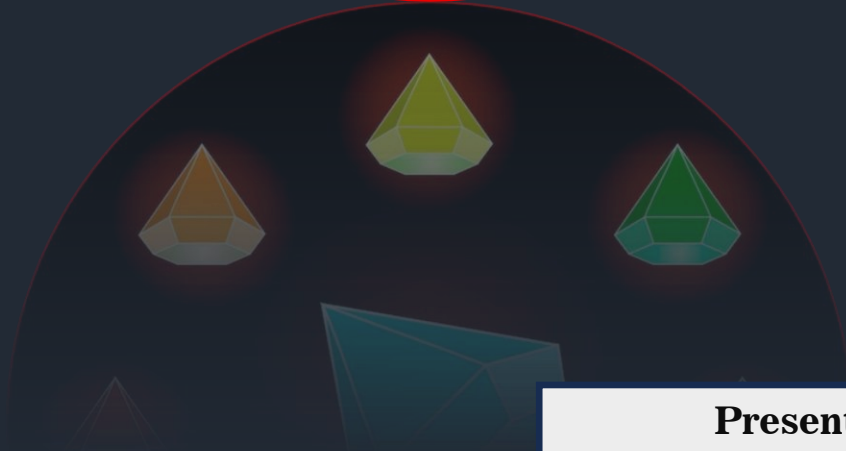


Final project report

GEM STONES PRICE PREDICTION



Group 09

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Abstract

This study focuses on predicting gemstone prices by combining a natural diamond sales dataset with a cubic zirconia dataset. First, we applied clustering methods such as FAMD, K-Means, and K-Prototypes, and evaluated clusters using validation metrics and UMAP visualization. Then we perform Exploratory Data Analysis (EDA) to understand the data patterns.

For price prediction, we fitted several models including Linear Regression, Lasso, Ridge, Elastic Net, Random Forest, Histogram Gradient Boosting, XGBoost, and CatBoost. After fine-tuning and Bayesian optimization, the CatBoost model gave the best performance.

We also analyzed feature importance and SHAP values. After removing less important variables (cut, table, and depth), the reduced model showed only a small drop in accuracy. Therefore, the full model is best for maximum accuracy, while the reduced model can be used for simpler implementation with slightly lower performance.

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Introduction

The Exploratory Data Analysis (EDA) phase of this diamond price prediction project provided a comprehensive understanding of the dataset by examining its structure, variable distributions, and relationships between diamond characteristics and price. Special attention was given to comparing real diamonds and artificial diamonds to identify differences in their physical properties and pricing behavior. These preliminary insights helped uncover key attributes such as carat, cut, clarity and other structural features that influence price variations.

Building upon these foundational findings, the Advanced Data Analysis phase focuses on developing predictive models to accurately estimate diamond prices while validating patterns identified during EDA. This phase further investigates the combined effects of physical attributes and diamond type (real vs. artificial) on price formation. Advanced analytical techniques such as dimensionality reduction, clustering, and machine learning models are applied to improve prediction accuracy and identify the most significant drivers of diamond pricing.

Description of the question

This study aims to support different stakeholders in the diamond market by providing insights into diamond pricing and quality differences between real and artificial diamonds. The analysis helps sellers maximize profits, assists buyers in selecting diamonds within their budget, and guides jewellery makers in choosing cost-effective materials while maintaining good quality.

1. What diamond features increase the price, and how can sellers use these features to sell real and artificial diamonds at higher prices? (for sellers)
2. How can buyers identify a good quality diamond within their budget based on the factors that affect price? (buyers)
3. How can jewellery makers choose between real and artificial diamonds to maintain good quality while reducing production costs? (jewellery makers)
4. Which machine learning approach (linear, distance-based, or tree-based ensemble) achieves the highest predictive accuracy after hyperparameter optimization?

Description of the dataset

Here we **combined two data sets** because both datasets represent the same market context (time period, seller, type, quality grading standards) to develop the pricing and recommendation logic. The two datasets are “**Diamonds sale dataset**” and “**Gemstone Price Prediction dataset**” which consider a gem made synthetically and low cost in a lab. Both datasets are taken from Kaggle website. Diamonds sale dataset contains 53940 observations and 9 variables and Gemstone Price Prediction dataset contains 26967 observations and 9 variables. Our combined dataset (final dataset) containing 80,072 observations after removing duplicates and unrealistic values and has 12 variables after doing feature engineering. Among them dataset has 4 categorical variables (Type, color, clarity and cut) and 8 numerical variables. The training dataset contain 64561 observation while testing dataset contain 16141 observations.

By combining the real diamond dataset and the artificial diamond(cubic zirconia which are made in the lab with low price than real diamond dataset, we can directly compare their features and prices in one analysis. This helps us clearly understand the differences and similarities between the two types of diamonds. It also improves the accuracy of our price prediction model because it learns from a larger and more diverse dataset.

Variable	Type	Description
LogPrice (target variable)	quantitative	Ln transformation of diamond prices
Price	quantitative	Price of the cubic zirconia/real diamond
X	quantitative	Length of cubic zirconia/real diamond in mm
Y	quantitative	Width of cubic zirconia/real diamond in mm
Z	quantitative	Height of cubic zirconia/real diamond in mm
Table	quantitative	How large the top surface of the stone is compared to its total width as a percentage.
Depth	quantitative	The Height of a cubic zirconia, measured from the Culet to the table, divided by its average Girdle Diameter.
Cut	qualitative	Describe the cut quality of the cubic zirconia/real diamond.
Color	qualitative	Color of the cubic zirconia/real diamond. With D being the best and J the worst.
Clarity	qualitative	Cubic zirconia/real diamond clarity refers to the absence of the Inclusions and Blemishes. . (In order from Best to Worst, FL = flawless, I3= level 3 inclusions) FL, IF, VVS1, VVS2, VS1, VS2, SI1, SI2, I1, I2, I3
Carat	quantitative	Carat weight of cubic zirconia/real diamond.
Type	qualitative	Type of the diamond (Zirconia or Diamond)

Table 1 : variable description

Data preprocessing

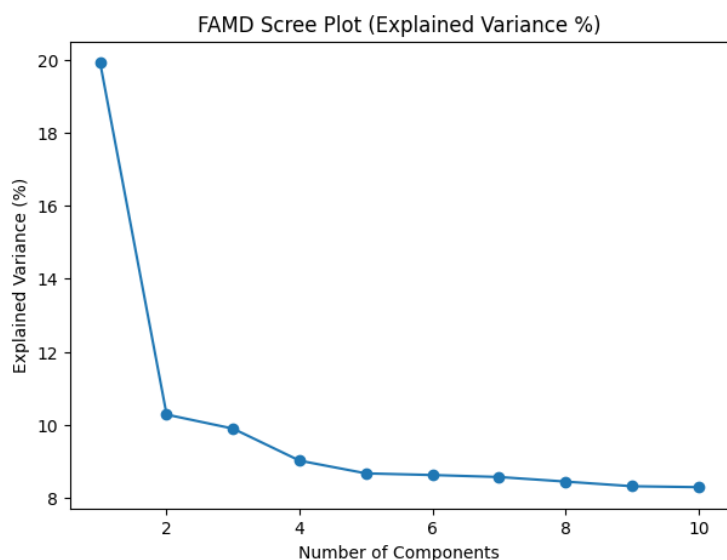
- We first checked duplicates and we found 355 duplicates. So we remove duplicates keeping first occurrence then out of 355 duplicate rows removed 180 rows.
- Our dataset has missing values only in one column (Depth). The training dataset has 567 rows of missing values in Depth column and testing dataset has 130 missing value rows in Depth column. So we fit a model using random forest for training dataset and predict missing values for Depth and we use same parameters and impute missing values for Depth in testing dataset.
- We have unrealistic values that means for height, weight and length we have zero values. So we remove those values from our dataset.
- We split dataset in to train (80%) and test (20%) having 64561 and 16141 observations respectively.

Feature Engineering

- We create Type column which does not exist in previous datasets to identify diamond type when combining the two datasets.
- We create LogPrice variable by getting Ln of Price column.
$$\text{LogPrice} = \ln(\text{Price})$$

This is our target variable because our prices in the two dataset has different currency types. Our real diamonds prices are in USD and cubic zirconia diamonds prices are in Indian Rupees. So we convert Indian rupees in to USD (**1 Indian Rupee = 0.011 US Dollar from Google**) by multiplying all Indian rupees prices from this amount, after that we have large scale differences . So we get Ln transformation of Price column to reduce the variation of scaling in prices. So we use LogPrice as our target variable.

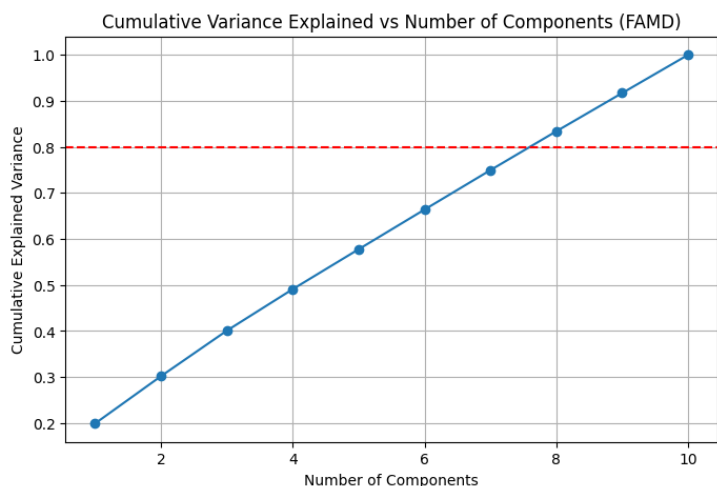
Dimensionality Reduction with FAMD (Factor Analysis of Mixed Data)



Factor Analysis of Mixed Data (FAMD) was applied to identify potential latent structures between categorical and numerical variables since our dataset has both categorical and numerical variables. This figure illustrates that variability Explained by each components in our dataset. We can see that first component explained around 20% variability of our dataset and from 4 to 10 variables variance is relatively evenly distributed.

Figure 1: FAMD scree plot

Explained Variance (%):
[19.92 10.28 9.89 9.02 8.67 8.62 8.57 8.44 8.31 8.29]



This plot shows that cumulative variance explained by the components in our dataset. This graph helps us to identify number of components which are explained more variability (around 80%) in our dataset. According to this graph we can see that 8 components are explained 80% of cumulative variance in our dataset. So from this, we can use 8 principal components which explained more variability in our dataset.

Figure 2: Cumulative variance explained plot

Cumulative Explained Variance (%):
[19.92 30.19 40.08 49.1 57.77 66.39 74.95 83.4 91.71 100.]

Cluster Analysis

Cluster analysis is a technique used to group similar data points together based on their characteristics. It helps to identify patterns or natural groupings within a dataset without using predefined labels. This method is useful because it allows us to understand hidden structures in the data, such as grouping diamonds with similar features or price ranges, which supports better decision making and analysis.

Different methods are applied to perform clustering, including **K-Means** for numerical data, **K-Prototypes** for mixed numerical and categorical data, and **UMAP combined with K-Means** for visualizing high dimensional data in clusters. Cluster evaluation metrics are then used to measure how well the data points are grouped, ensuring meaningful and accurate clusters.

K means clustering for FAMD data

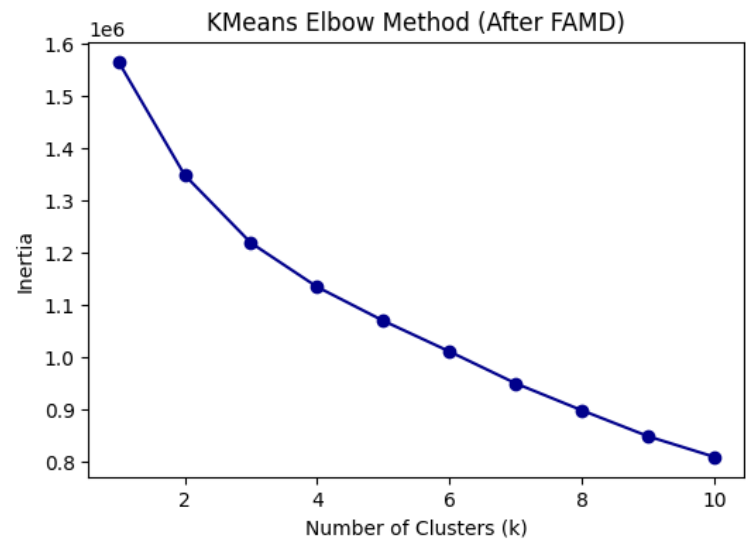
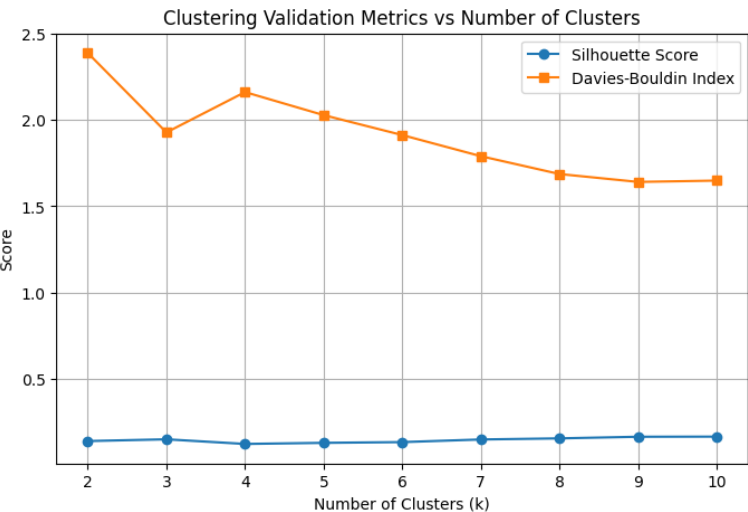


Figure 3: K means elbow plot

The elbow method was first applied to determine the optimal number of clusters for K-Means. The analysis examined K values ranging from 2 to 11. However, k means is only applicable for numerical data and since our dataset has both numerical and categorical variables we first do FAMD and after that we use the k means method. From this figure we can see that there is no clear elbow point which guarantees there is no clear cluster structure.

Cluster validation metrics



The Silhouette scores are consistently low (0.13–0.17) across all values of k, indicating weak separation and high overlap between clusters. There is no clear peak suggesting an optimal number of clusters. DBI decreases slowly as k increases, without a sharp drop or strong minimum. This suggests there is **no strong natural cluster structure**, and increasing k is likely creating artificial partitions rather than meaningful clusters. So **this figure illustrates that there is no clear cluster structure in our dataset.**

Figure 4: Cluster validation metrics - Silhouette and Davies-Bouldin Index plots

k = 2	Silhouette Score: 0.1384	Davies-Bouldin Index: 2.3885
k = 3	Silhouette Score: 0.1486	Davies-Bouldin Index: 1.9279
k = 4	Silhouette Score: 0.1218	Davies-Bouldin Index: 2.1619
k = 5	Silhouette Score: 0.1281	Davies-Bouldin Index: 2.0282
k = 6	Silhouette Score: 0.1324	Davies-Bouldin Index: 1.9126
k = 7	Silhouette Score: 0.1477	Davies-Bouldin Index: 1.7901
k = 8	Silhouette Score: 0.1540	Davies-Bouldin Index: 1.6867
k = 9	Silhouette Score: 0.1634	Davies-Bouldin Index: 1.6409
k = 10	Silhouette Score: 0.1642	Davies-Bouldin Index: 1.6487

K-Prototype Clustering with Gower Distance

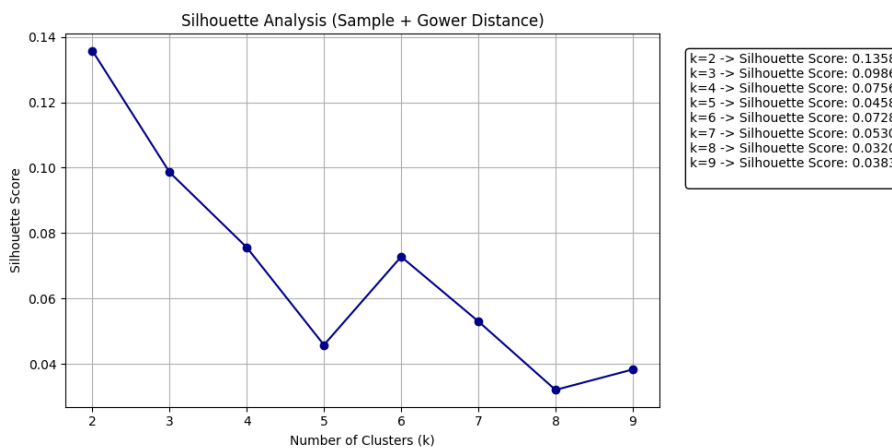
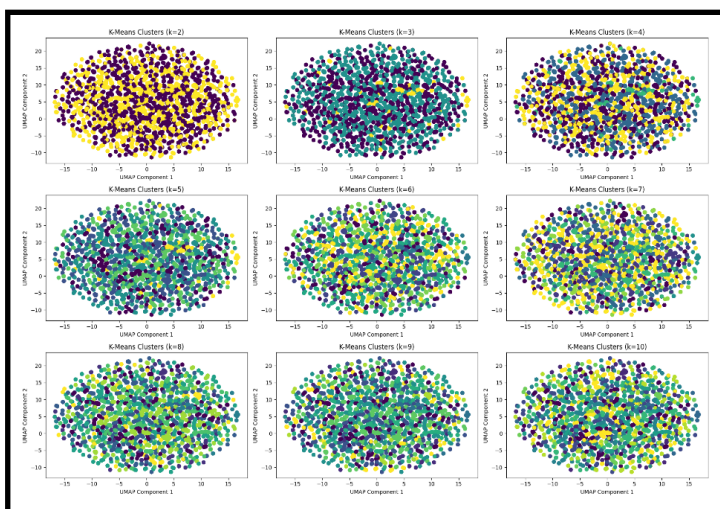


Figure 5: K-Prototype Clustering Results with Gower Distance

K-Prototype clustering was applied directly to the original dataset using Gower distance, which is well suited for mixed data containing both numerical and categorical variables. By working in the full feature space rather than on FAMD components, this approach avoids potential information loss caused by dimensionality reduction. The algorithm was evaluated across multiple values of K to determine whether any natural cluster structure exists in the data. **As a conclusion from this figure we can get that no clear or meaningful clusters emerged.**

UMAP visualization



UMAP (Uniform Manifold Approximation and Projection) is a dimensionality reduction technique that projects high dimensional data into a lower dimensional space (usually 2D or 3D) while preserving the underlying structure of the data as much as possible. UMAP visualization is important because many datasets exist in dozens or hundreds of dimensions, which humans cannot directly interpret. By projecting the data into two dimensions, UMAP allows us to visually inspect patterns, groupings, density regions, and potential cluster separation before or after applying clustering algorithms like K-means.

Figure 6: K-Means Cluster Visualization on UMAP Components for K=2 to K=10

In the provided plots, K-means clustering results for different values of k (from 2 to 10) are visualized on the same UMAP projection. Across all values of k , the data points form a roughly continuous, oval-shaped cloud without clearly separated dense regions or natural gaps between groups. The cluster colors appear heavily intermixed rather than forming distinct, well-separated regions. As k increases, the algorithm simply partitions the continuous space into more segments, but these segments do not correspond to visually distinct natural clusters.

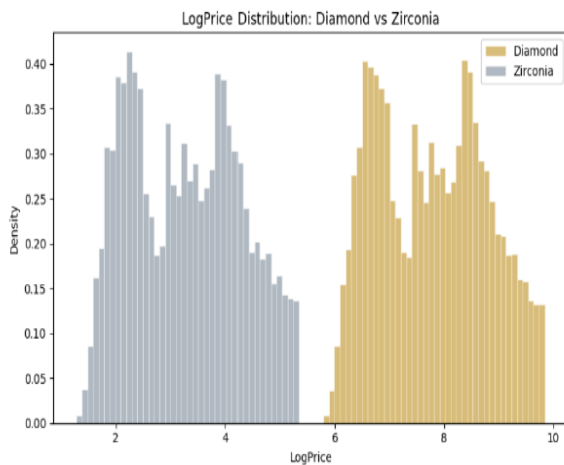
- **Based on the UMAP visualization, there is no strong visual evidence of clearly separable clusters in the data. The structure appears continuous rather than grouped into distinct regions. Therefore, the dataset likely does not contain well defined natural clusters.**

✓ **All these methods guarantee that our dataset does not contain natural, clear separated cluster structure.**

Descriptive Analysis Findings

Univariate

Target Variable + log transform



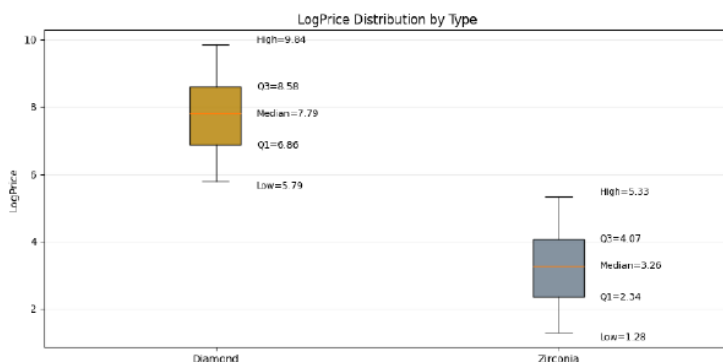
TEST 1: Diamond vs Zirconia difference in LogPrice
 H0: Diamond and Zirconia LogPrice distributions are the same
 H1: Distributions differ
 alpha = 0.05
 Mann-Whitney U statistic = 9.26e+08, p-value = 0
 Cliff's delta = 1 (near ± 1 = huge effect)
 Decision: Reject H0
 Conclusion: Type has a statistically significant effect on LogPrice.
 Implication: strongly supports separate modeling or using Type as a major feature.

Figure 7: Target Variable distribution

Price is heavily right skewed (many cheap items, few very expensive), so raw Price isn't close to normal and variance will grow with price. Log(Price) improves it (more balanced, more linear-friendly), but it's still not normal. So LogPrice distribution is still skewed and variance between Diamond and Zirconia is very different so we use Mann-Whitney test (non parametric test) which is used when normality assumption is violated to validate the target variable distribution but since we have large sample sizes we can here use t test even if the normality assumption is violated.

Bivariate

1) LogPrice vs Type



===== Independent Samples t-Test =====

H0: Mean log price of Diamond = Mean log price of Zirconia
 H1: Means are different

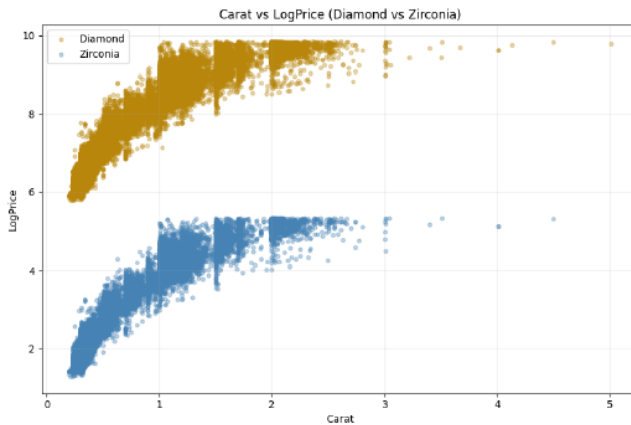
T-statistic: 532.1925
 P-value: 0.0

Decision: Reject the Null Hypothesis
 Conclusion: There is a statistically significant difference
 Mean log price differs between Diamond and Zirconia.

Figure 8: LogPrice distribution by Type

This figure illustrates that LogPrice of Diamond is much higher than Zirconia across the entire distribution because of big separation in medians. So as conclusion we can say that type is dominant driver of LogPrice. We used an independent t-test to compare LogPrice between Type groups because our sample size is large, and with large samples the t-test is robust to mild violations of normality due to the Central Limit Theorem. Our hypothesis was about comparing the mean LogPrice between the two types. Therefore, the independent t-test was used to determine whether the average LogPrice differs significantly between the groups.

2) LogPrice vs carat



TEST: Type effect on LogPrice controlling for Carat
H0: No difference in expected LogPrice between types at same Carat ($\beta_{\text{Type}}=0$)
H1: Difference exists ($\beta_{\text{Type}} \neq 0$)
coef (zirconia - diamond) = -4.514730470183454
p-value = 0.0
Decision: REJECT H0
Approx price ratio zirconia/diamond = 0.010946555152255

Figure 9: LogPrice vs carat

The plot shows Carat vs LogPrice for two stone types. Here we can see strong positive relationship indicating for both types price increases as carat increases. Nonlinear patterns can be seen there, the increases steeper at low carats and then flattens somewhat at higher carats. A few points at higher carat don't follow the main cloud indicating potential unusual stones or data outliers.

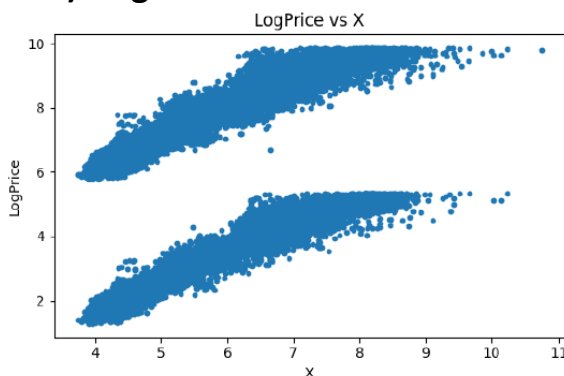
Here we use ANCOVA test (Analysis of Covariance) as statistical test to validate this finding because here we want to test after controlling carat is there any difference between Types.

So, $\text{LogPrice} = \beta_0 + \beta_1(\text{Carat}) + \beta_2(\text{Type})$.

Coefficient (zirconia - diamond) = -4.5147 means that at the same carat weight, zirconia's LogPrice is 4.51 units lower than diamond's.

Since this is on the log scale, we exponentiate to interpret as a price ratio, $e^{-4.5147} \approx 0.0109$. So we can say that Zirconia costs about 1.1% of the price of a diamond of the same carat.

3) LogPrice vs X



--- LogPrice vs X (Spearman) ---
H0: $\rho = 0$ (no monotonic association)
H1: $\rho \neq 0$ (monotonic association exists)
 $\rho = 0.5391$, p-value = 0
Decision: REJECT H0

Figure 10: LogPrice vs X

Strong positive relationship indicates for both materials, LogPrice increases as X increases. Two distinct bands suggest different groups (Type). Spearman correlation was used because the relationship between LogPrice and X appears monotonic but not strictly linear. Additionally, the presence of outliers makes Spearman more appropriate than Pearson correlation.

4) LogPrice vs Y

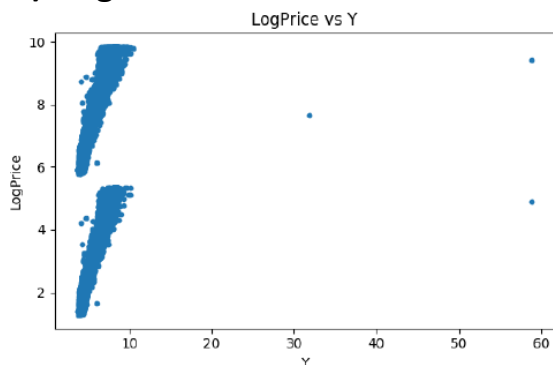


Figure 11: Price vs Y

--- LogPrice vs Y (Spearman) ---
 H0: $\rho = 0$ (no monotonic association)
 H1: $\rho \neq 0$ (monotonic association exists)
 $\rho = 0.5389$, p-value = 0
 Decision: REJECT H0

Strong positive relationship indicates for both materials, LogPrice increases as increases which can say bigger stones higher price. Here we can see few outliers also. Spearman correlation was used because the relationship between LogPrice and Y appears monotonic but not strictly linear. Additionally, the presence of outliers makes Spearman more appropriate than Pearson correlation.

5) LogPrice vs Z

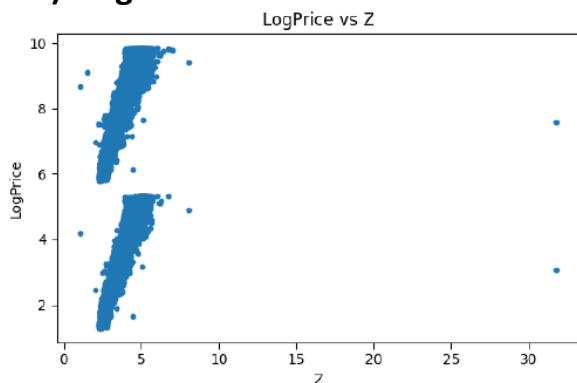


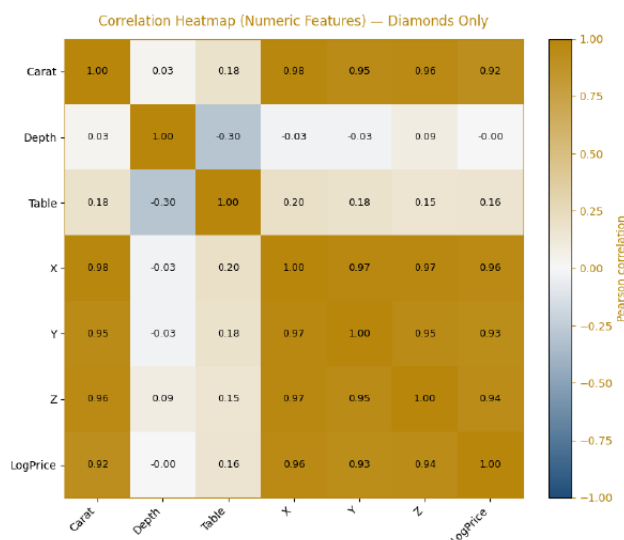
Figure 12: Price vs Z

--- LogPrice vs Z (Spearman) ---
 H0: $\rho = 0$ (no monotonic association)
 H1: $\rho \neq 0$ (monotonic association exists)
 $\rho = 0.5366$, p-value = 0
 Decision: REJECT H0

Strong positive relationship indicates for both materials, LogPrice increases as Z increases. Here we see few outliers also. Here we also use Spearman correlation test to validate this finding because here we can see monotonic relationship.

- ❖ From here we use only diamonds to identify what variables influence for price because zirconia and diamond have same distribution because zirconia price is only an engineered scaled version so comparing both may be hidden patterns and make the trend line

Correlation heat map for numerical variables (Diamonds only)



Carat is the strongest driver of LogPrice indicating strong positive relationship. Dimensions are highly correlated with carat indicating that bigger stones have larger measurements. Dimensions also correlate strongly with LogPrice. There can see strong multicollinearity among dimensions indicating there can be overlapping information. Depth vs Table is moderately negative indicating with larger table tend to have slightly lower depth.

Figure 13: Correlation heat map for diamonds

6) LogPrice by Cut

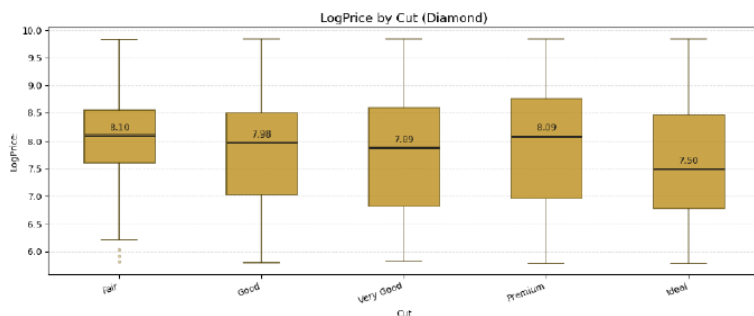


Figure 14: LogPrice by Cut

Cut alone doesnot indicate influence for price strongly. Premium (and Fair) have the highest medians in this plot indicating highest typical price. Lowest typical prices appear in Ideal (best cut) shows the lowest median. Fair has a few low price outliers.

7) LogPrice by Color

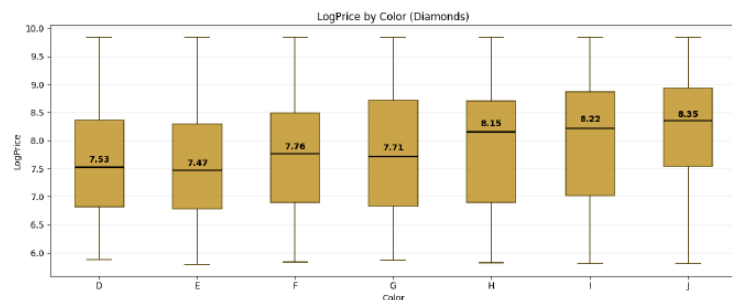


Figure 15: LogPrice by Color

This figure shows us that the median LogPrice generally increases from D(best) to J (worst), meaning lower color grades tend to have higher prices in this dataset which is not suitable for real world. Each color group has a wide IQR range imply that prices vary a lot within the same color grade. The boxes and whiskers overlap heavily across colors, so color alone doesn't cleanly separate price.

8) LogPrice by Clarity

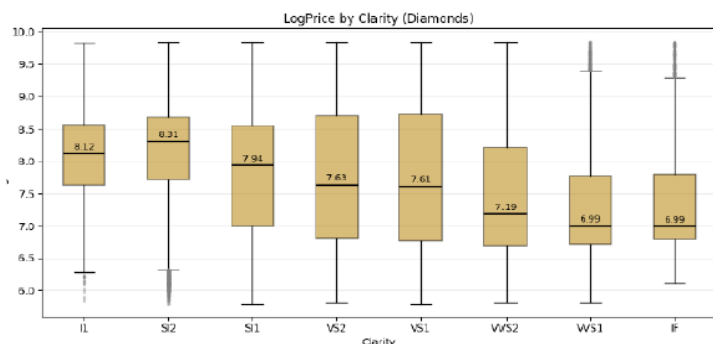


Figure 16: LogPrice by Clarity

This graph shows LogPrice for diamonds grouped by clarity. Prices tend to be lower at higher clarity grades in our dataset. That's not saying "better clarity is cheaper" in reality, it's a sign that clarity is confounded with carat/size here. This may be due to high clarity stones in our dataset may often be smaller. This plot suggests clarity alone doesn't explain price well unless control for carat. A few outliers are appear in higher clarity levels and lower clarity levels. That is may be due to some higher clarity stones have large carat weight and some worst clarity stones have small carat weight respectively.

Our EDA findings

- ✓ carat directly increase diamond price
- ✓ Type is a dominant driver of price
- ✓ Dimensions (X,Y,Z) increases diamonds price
- ✓ High clarity stones decreases diamond price which is unrealistic for rael world and we can say this can be happen due to the high clarity stones get smaller carat weight.

Advanced Analysis

In model evaluation, we compare training and testing performance using metrics such as MSE (lower is better), R^2 (higher is better), and MAPE (lower is better). This helps assess the model's generalization ability. If the training performance is much better than the testing performance (large gap), the model is overfitting, meaning it memorizes training data but fails on new data. If both training and testing performances are poor, the model is underfitting, meaning it fails to capture the underlying patterns. The ideal situation is achieving strong test performance with a small and reasonable gap between train and test results, indicating good generalization.

Fitted models

Model	Train_R2	Test_R2	Train_MSE	Test_MSE	Train_MAPE(%)	Test_MAPE(%)
CatBoost	0.999104	0.998882	0.004994	0.006182	0.998067	1.102665
XGBoost	0.999049	0.998835	0.005300	0.006444	1.030571	1.144645
RandomForest	0.999783	0.998515	0.001208	0.008212	0.488573	1.296243
HistGradientBoosting	0.998422	0.998395	0.008793	0.008875	1.396531	1.423879
Ridge	0.996588	0.996651	0.019014	0.019014	2.109776	2.102266
LinearRegression	0.996588	0.996651	0.019014	0.018518	2.109750	2.102241
ElasticNet	0.996548	0.996624	0.019234	0.018671	2.121337	2.113961
Lasso	0.996507	0.019465	0.019465	0.018671	2.131287	2.125186

Table 2 : Fitted model comparison metrics

Why CatBoost stands out (Baseline Model)

- It performs very well on test data, especially when the dataset contains both numeric and categorical variables. It handles categorical features in a smart and stable way, better than simple one-hot encoding in many cases. Compared to all other baseline models, it has the highest Train and Test R^2 values and the lowest MSE values, which means it gives more accurate predictions with smaller errors.

Here although the random forest model has highest R^2 values and lowest MSE values it perform overfitting results which means training R^2 is higher than test R^2 , training MSE is smaller than testing MSE and same as MAPE values. So we ignore Random Forest model as our best model.

However, after selecting the best baseline models, we further tuned **XGBoost**, **CatBoost**, and **HistGradientBoosting** to improve their performance even more.

Feedback : It is better to tuned above all fitted models in Table 2 and then choose the best model than tuned only XGBoost, CatBoost and HistGradientBoost and choose the best model among them because after tuning all those models sometimes some models less perform earlier may be perform better after tuning.

Fine tuned models

Model	Version	Train_R2	Train_MSE	Train_MAPE(%)	Test_R2	Test_MSE	Test_MAPE(%)
XGBoost	Tuned	0.999612	0.002160	0.631571	0.999066	0.005164	0.984758
CatBoost	Tuned	0.999387	0.003414	0.814332	0.998993	0.005567	1.033080
CatBoost	Baseline	0.999104	0.004994	0.998067	0.998882	0.006182	1.102665
XGBoost	Baseline	0.999049	0.005300	1.030571	0.998835	0.006444	1.144645
HistGradientBoosting	Tuned	0.998823	0.006556	1.157901	0.998648	0.007475	1.248487
HistGradientBoosting	Baseline	0.998422	0.008793	1.396531	0.998395	0.008875	1.423879

Table 3: Fine tuned model comparison matrices

XGBoost showed the greatest improvement from targeted hyperparameter tuning, with lower MSE and MAPE indicating more accurate predictions on unseen data. CatBoost remains a strong candidate and could benefit from further optimization. So we optimize using bayesian optimization.

Bayesian optimization smartly guesses which parameter settings to try next by learning from past results, so it finds the best settings with fewer trials and less wasted time.

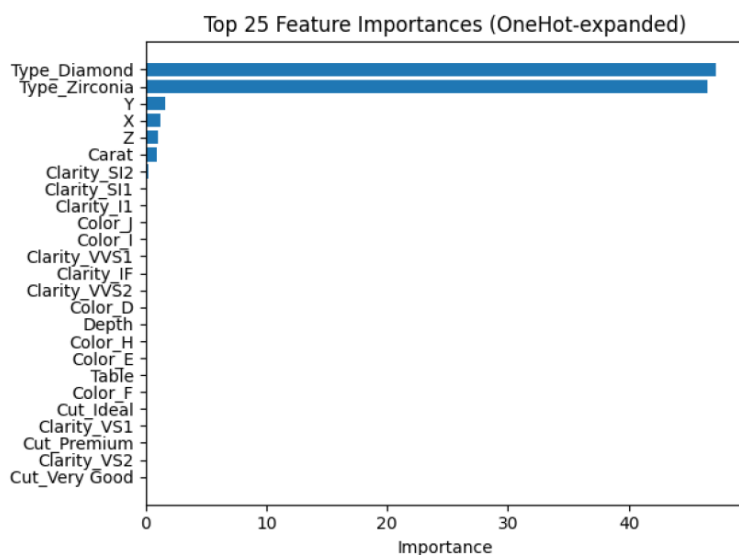
Bayesian Optimization results

Model	Version	Train_R2	Test_R2	Train_MSE	Test_MSE	Train_MAPE(%)	Test_MAPE(%)
CatBoost	Baseline	0.999104	0.998882	0.004994	0.006182	0.998067	1.102665
CatBoost	Tuned	0.999387	0.998993	0.003414	0.005567	0.814332	1.033080
CatBoost	Tuned (Bayesian)	0.999695	0.999187	0.001697	0.004497	0.551630	0.867376
XGBoost	Baseline	0.999049	0.998835	0.005300	0.006444	1.030571	1.144645
XGBoost	Tuned	0.999612	0.999066	0.002160	0.005164	0.631571	0.984758
XGBoost	Tuned (Bayesian)	0.999278	0.998913	0.004021	0.006012	0.885593	1.101673
HistGradientBoosting	Baseline	0.998422	0.998395	0.008793	0.008875	1.396531	1.423879
HistGradientBoosting	Tuned	0.998823	0.998648	0.006556	0.007475	1.157901	1.248487
HistGradientBoosting	Tuned (Bayesian)	0.998784	0.998654	0.006775	0.007445	1.180706	1.252677

Table 4: Bayesian tuned model comparison matrices

From the results of bayesian optimization we can choose **CatBoost** as the **final best model** which having highest R2 and lowest MSE and MAPE among XGBoost and HistGradientBoosting models.

Feature importance plot (one hot encoded variables) of CatBoost Bayesian Tunned Model

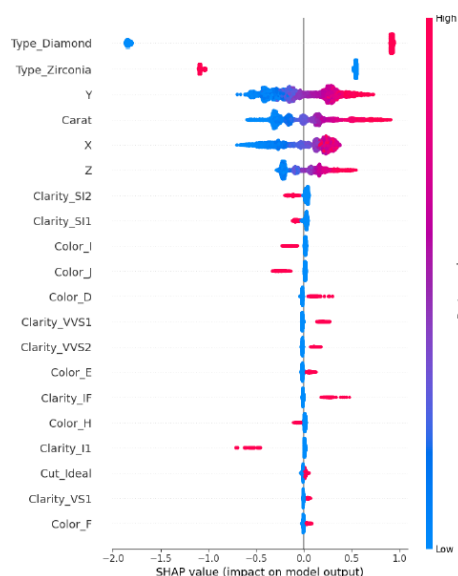


The feature importance results of one hot encoded variables show that **stone type (Diamond vs Zirconia)** is by far the most influential variable in the model, contributing significantly more than any other feature. This indicates that the model's predictions are driven primarily by whether the gem is a diamond or not. In comparison, physical characteristics such as dimensions (X, Y, Z) and Carat have only minor influence, while quality related attributes like Clarity, Color, Cut, Depth, and Table contribute very little individually.

Figure 17: Feature Importance plot of one hot encoded features

- ✓ **Feedback :** However the Type_Diamond and Type_Zirconia both check whether the stone is diamond or not, but in importance plot Type_Diamond and Type_Zirconia both have different importance although they check same thing. We can say that this is because model prefer to choose one variable among them very much than the other, and this may not be actual importance. So to overcome this problem we plot importance plot for our original variables and see the importance of each feature to target variable.

Shap values of one hot encoded feature importance plot



The SHAP summary plot shows that the most important factor influencing price prediction is the type of stone, where diamonds strongly increase the predicted price and zirconia significantly decreases it. Among all numerical features, carat weight and the physical dimensions (X, Y, and Z) have a major positive impact, meaning larger and heavier stones lead to higher predicted prices. Positive SHAP values indicate an increase in price, while negative values indicate a decrease. Although clarity grades such as IF and VVS1, VVS2 contribute positively, and color has some effect, their influence is relatively smaller compared to stone type and size. Overall, the model primarily relies on stone type and physical size characteristics, with clarity and color acting as secondary factors in price determination.

Figure 18: Shap values summary plot of one hot encoded features

Importance aggregated to original variables

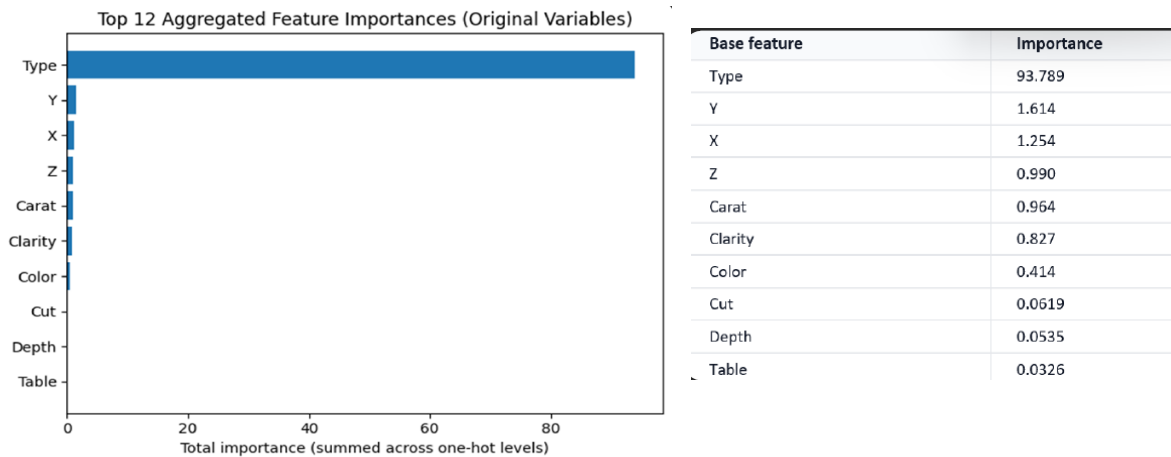


Figure 19: Aggregated feature importance plot of original variables

This graph shows that **Type** is the most important factor in predicting price. The model mostly looks at whether the stone is a diamond or zirconia when deciding the price. Other features like **carat** and the stone's size (X, Y, Z) also affect the price, but much less than the type. Features such as **clarity, color, cut, depth, and table** have only a small influence. In simple words, the model mainly depends on the stone type, then size, and gives very little importance to the other features.

We can see importance plot Cut / Depth / Table are very low impact in this model. So we tested a reduced model removing them to see if we could:

- Simplify the feature set
- Reduce noise / multicollinearity risk
- Speed up Prediction

Multicollinearity among numerical variables

Feature		VIF
0	const	4835.577800
1	Carat	24.494068
2	Depth	1.429970
3	Table	1.143925
4	X	50.048079
5	Y	14.418559
6	Z	19.022602

Among numerical variables when we check VIF values we can see that Carat and Dimensions (X,Y,Z) have high multicollinearity while Depth and Table perform less multicollinearity.

Why we try reduced model ?

- ✓ Simpler model can be easier to maintain
- ✓ Fewer inputs can reduce data collection burden
- ✓ Sometimes improves robustness by removing noise

Model	Set up	Test R2	Test MSE	Test MAPE
CatBoost	Full (Bayes Tunned)	0.999187	0.004497	0.8674
CatBoost	Reduced	0.998987	0.005602	0.9923
XGBoost	Reduced	0.998764	0.006835	1.1618

Table 5: Model comparison with full model and reduced model

- According to this we keep our full model for maximum accuracy.
- But We can use the Reduce model if the accuracy drop not that much a problem.

Conclusion

This study analyzed and predicted gemstone prices by combining real diamond and cubic zirconia datasets. Exploratory Data Analysis showed that stone type and carat weight are the most influential factors, with dimensions (X, Y, Z) also positively affecting price. Quality related attributes such as clarity, color, cut, depth, and table have relatively smaller effects once size and type are considered. Cluster analysis using FAMD, K-Means, K-Prototype, and UMAP revealed no clear natural cluster structure, indicating that the data follows a continuous pattern rather than distinct groups.

Among all predictive models tested, CatBoost with Bayesian optimization achieved the best performance, showing the highest R^2 and lowest MSE and MAPE on the test data. Feature importance and SHAP analysis confirmed that stone type is the dominant predictor, followed by carat and physical dimensions. A reduced model excluding cut, depth, and table showed only a small drop in accuracy, meaning it can be used when a simpler model is preferred.

Overall, this study demonstrates that advanced ensemble learning methods, particularly CatBoost with Bayesian tuning, provide highly accurate gemstone price predictions. The findings offer valuable insights for sellers, buyers, and jewellery manufacturers by clearly identifying the most influential pricing factors and providing a reliable predictive framework for decision making.

Appendix

Link for our datasets :

Diamond sale dataset –

<https://www.kaggle.com/datasets/sahilnbajaj/diamonds-sale-data/data>

Cubic Zirconia dataset –

https://www.kaggle.com/datasets/colearninglounge/gemstone-price-prediction?select=cubic_zirconia.csv

Resources :

<https://www.brilliantearth.com/diamond/buying-guide/carat/>

<https://www.kaggle.com/code/karnikakapoor/diamond-price-prediction>

Python codes:

Pre processing :

<https://colab.research.google.com/drive/1tsbGiC72ISRJlex8SIbh1wEmwobDDEku?usp=sharing>

Clustering:

<https://colab.research.google.com/drive/1DBYKjsRJ5NujR4m2lQb8LVvDgPawRcze?usp=sharing>

EDA :

https://colab.research.google.com/drive/193kOZLB_PUz6SckdKd8HtiUtJoSdIXDS?usp=sharing

Model fitting :

<https://colab.research.google.com/drive/1QQ9Qvhw345jwNayR8dKntdwSb-Wuj5FF?usp=sharing>